Radar Cross Section in DiscoGMPI: Theory, Conventions, and Numerical Evaluation

DiscoGMPI Documentation

July 17, 2026

Abstract

This document gives a self-contained description of radar cross section (RCS), the time-harmonic scattering model used for perfectly electrically conducting targets, and the numerical procedure implemented in DiscoGMPI. It fixes the physical assumptions, phase convention, angular convention, normalization, near-to-far transformation, time-windowed Fourier extraction, DG surface quadrature, MPI reduction, output files, and validation strategy. The focus is the metallic-sphere workflow implemented in `examples/distributed_metallic_sphere_scattering.jl`, with validation and Mie-series comparison through `examples/validate_coarse_sphere_rcs.jl` and `examples/postprocess_metallic_sphere_mie_rcs.py`. The document also describes the NASA/Austin PEC Almond benchmark workflow implemented in `examples/distributed_pec_almond_scatte` and `examples/postprocess_pec_almond_rcs.py`.

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1 Purpose of RCS

Radar cross section is a far-field measure of how strongly a target scatters an incident electromagnetic wave into a selected observation direction. It is defined so that a distant receiver can interpret the scattered field as if it had been produced by an equivalent isotropic scatterer with area σ. Large RCS means that the object returns a strong scattered field in that direction; small RCS means that it scatters weakly.

For a monochromatic incident plane wave with electric-field amplitude $|\boldsymbol{E}_{\mathrm{inc}}|$, the far scattered field has the asymptotic form

$$\widehat{\boldsymbol{E}_{\mathrm{scat}}}(\boldsymbol{x}) = \frac{\mathrm{e}^{\mathrm{i}kr}}{r}\boldsymbol{E}^{\infty}(\widehat{\boldsymbol{r}}) + \mathcal{O}(r^{-2}), \qquad r = |\boldsymbol{x}|, \qquad \widehat{\boldsymbol{r}} = \frac{\boldsymbol{x}}{|\boldsymbol{x}|}. \tag{1}$$

The RCS in direction $\widehat{\boldsymbol{r}}$ is

$$\sigma(\widehat{\boldsymbol{r}}) = \lim_{r\to\infty} 4\pi r^2 \frac{|\widehat{\boldsymbol{E}_{\mathrm{scat}}}(r\widehat{\boldsymbol{r}})|^2}{|\widehat{\boldsymbol{E}_{\mathrm{inc}}}|^2} = 4\pi \frac{|\boldsymbol{E}^{\infty}(\widehat{\boldsymbol{r}})|^2}{|\widehat{\boldsymbol{E}_{\mathrm{inc}}}|^2}. \tag{2}$$

In DiscoGMPI, the incident amplitude is denoted by A, so $|\widehat{\boldsymbol{E}_{\mathrm{inc}}}| = A$. For a sphere of radius a, postprocessing usually plots the dimensionless normalized RCS

$$\frac{\sigma}{\pi a^2} \tag{3}$$

or its decibel value

$$10\log_{10}\left(\frac{\sigma}{\pi a^2}\right). \tag{4}$$

2 Physical Assumptions

The current DiscoGMPI metallic-sphere RCS workflow assumes:

1. The exterior medium is homogeneous, isotropic, and lossless outside the artificial absorbing layer.
2. The material parameters are scalar constants $\epsilon > 0$ and $\mu > 0$, with wave speed

$$c = \frac{1}{\sqrt{\epsilon\mu}}, \qquad Z = \sqrt{\frac{\mu}{\epsilon}},$$

   where Z is the wave impedance.
3. The scatterer is a perfectly electrically conducting (PEC) sphere centered in a Cartesian computational box.
4. The numerical unknown is the scattered field. The physical field is reconstructed as incident plus scattered field.
5. The outer computational box is truncated with the nonlinear Cartesian PML available in DiscoGMPI.
6. The RCS diagnostic is extracted from the PEC surface current. The current implementation does not yet use a Huygens box or volume near-to-far transformation.
7. The incident wave is monochromatic and the reported RCS is the Fourier component at the incident frequency. Time-domain transients must be excluded by choosing an appropriate `-rcs-start-time`.

3 Time-Harmonic Convention

DiscoGMPI uses the real incident wave

$$\boldsymbol{E}_{\mathrm{inc}}(\boldsymbol{x}, t) = A\,\widehat{\boldsymbol{x}}\cos(kz - \omega t) \tag{5}$$

for the default sphere experiment. This is equivalent to the phasor convention

$$\boldsymbol{E}(\boldsymbol{x}, t) = \mathrm{Re}\left\{\widehat{\boldsymbol{E}}(\boldsymbol{x})\,\mathrm{e}^{-\mathrm{i}\omega t}\right\}. \tag{6}$$

With this convention, the default incident phasors are

$$\widehat{\boldsymbol{E}_{\mathrm{inc}}}(\boldsymbol{x}) = A\,\widehat{\boldsymbol{x}}\,\mathrm{e}^{\mathrm{i}kz}, \qquad \widehat{\boldsymbol{H}_{\mathrm{inc}}}(\boldsymbol{x}) = \frac{A}{Z}\,\widehat{\boldsymbol{y}}\,\mathrm{e}^{\mathrm{i}kz}. \tag{7}$$

The corresponding real magnetic field is

$$\boldsymbol{H}_{\mathrm{inc}}(\boldsymbol{x}, t) = \frac{A}{Z}\,\widehat{\boldsymbol{y}}\cos(kz - \omega t). \tag{8}$$

The general plane-wave form is

$$\widehat{\boldsymbol{E}_{\mathrm{inc}}}(\boldsymbol{x}) = A\,\boldsymbol{p}\,\mathrm{e}^{\mathrm{i}k\boldsymbol{d}\cdot\boldsymbol{x}}, \qquad \widehat{\boldsymbol{H}_{\mathrm{inc}}}(\boldsymbol{x}) = \frac{A}{Z}\,(\boldsymbol{d}\times\boldsymbol{p})\,\mathrm{e}^{\mathrm{i}k\boldsymbol{d}\cdot\boldsymbol{x}}, \tag{9}$$

where $\boldsymbol{d}$ is the unit propagation direction, $\boldsymbol{p}$ is the unit electric polarization, and $\boldsymbol{d}\cdot\boldsymbol{p} = 0$. The default code values are

$$\boldsymbol{d} = \widehat{\boldsymbol{z}}, \qquad \boldsymbol{p} = \widehat{\boldsymbol{x}}.$$

4 Maxwell Scattering Problem

In the physical exterior domain Ω^+, the total fields satisfy

$$\epsilon \frac{\partial \boldsymbol{E}_{\mathrm{tot}}}{\partial t} = \nabla \times \boldsymbol{H}_{\mathrm{tot}}, \tag{10}$$

$$\mu \frac{\partial \boldsymbol{H}_{\mathrm{tot}}}{\partial t} = -\nabla \times \boldsymbol{E}_{\mathrm{tot}}. \tag{11}$$

In phasor form, using $\mathrm{e}^{-\mathrm{i}\omega t}$,

$$-\mathrm{i}\omega\epsilon\, \widehat{\boldsymbol{E}_{\mathrm{tot}}} = \nabla \times \widehat{\boldsymbol{H}_{\mathrm{tot}}}, \tag{12}$$

$$-\mathrm{i}\omega\mu\, \widehat{\boldsymbol{H}_{\mathrm{tot}}} = -\nabla \times \widehat{\boldsymbol{E}_{\mathrm{tot}}}. \tag{13}$$

The wavenumber and angular frequency are related by

$$k = \frac{\omega}{c} = \omega\sqrt{\epsilon\mu} = \frac{2\pi}{\lambda}. \tag{14}$$

The total field is decomposed as

$$\boldsymbol{E}_{\mathrm{tot}} = \boldsymbol{E}_{\mathrm{inc}} + \boldsymbol{E}_{\mathrm{scat}}, \qquad \boldsymbol{H}_{\mathrm{tot}} = \boldsymbol{H}_{\mathrm{inc}} + \boldsymbol{H}_{\mathrm{scat}}. \tag{15}$$

The numerical field stored by the driver is

$$\boldsymbol{U}_h = \left(\boldsymbol{E}_{\mathrm{scat},h}, \boldsymbol{H}_{\mathrm{scat},h}\right). \tag{16}$$

This is useful because the incident field is known analytically everywhere, including on the metallic surface and in the PML region.

4.1 PEC Boundary Condition

On the metallic sphere Γ_{PEC}, the total tangential electric field must vanish:

$$\boldsymbol{n} \times \boldsymbol{E}_{\mathrm{tot}} = \boldsymbol{0} \qquad \text{on } \Gamma_{\mathrm{PEC}}. \tag{17}$$

Using the scattered-field decomposition, the condition imposed by the code is

$$\boldsymbol{n} \times \boldsymbol{E}_{\mathrm{scat}} = -\boldsymbol{n} \times \boldsymbol{E}_{\mathrm{inc}} \qquad \text{on } \Gamma_{\mathrm{PEC}}. \tag{18}$$

Here $\boldsymbol{n}$ is the normal pointing out of the PEC object into the exterior medium. The DG mesh is a box with an inner spherical hole; the geometric boundary normal stored for the exterior computational domain points opposite to the PEC-object normal on the inner sphere. Therefore `build_pec_sphere_face_samples` flips the boundary-face normal before using it in the PEC residual and RCS current:

$$\boldsymbol{n}_{\mathrm{RCS}} = -\boldsymbol{n}_{\mathrm{DG,inner}}. \tag{19}$$

4.2 Absorbing Layer

The physical scattering problem is posed in an unbounded exterior domain. The production driver truncates the domain with a Cartesian nonlinear PML near the outer box boundaries. The RCS diagnostic is not evaluated on the outer PML; it is evaluated on the PEC sphere surface. A reliable RCS run therefore requires the PML to suppress outgoing scattered waves without reflecting them back to the sphere during the Fourier accumulation window.

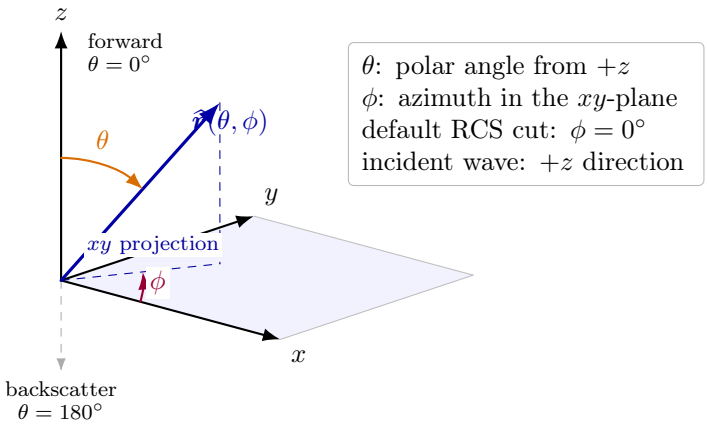


Figure 1: DiscoGMPI angular convention for RCS observation directions. The polar angle θ is measured from the $+z$ axis, while the azimuth ϕ is measured in the xy-plane from $+x$ toward $+y$.

5 Angular Coordinates

DiscoGMPI uses the standard spherical convention

$$\widehat{\boldsymbol{r}}(\theta,\phi) = (\sin\theta\cos\phi,\ \sin\theta\sin\phi,\ \cos\theta)\,, \tag{20}$$

with

$$0 \le \theta \le \pi, \qquad 0 \le \phi < 2\pi.$$

Here θ is the polar scattering angle measured away from the positive z-axis, and ϕ is the azimuthal angle in the xy-plane measured from the positive x-axis toward the positive y-axis. The convention is shown in Figure 1. For the default incident wave propagating in $+z$, $\theta = 0°$ is forward scattering and $\theta = 180°$ is backscatter.

The transverse basis vectors are

$$\widehat{\boldsymbol{e}}_\theta = (\cos\theta\cos\phi,\ \cos\theta\sin\phi,\ -\sin\theta)\,, \tag{21}$$

$$\widehat{\boldsymbol{e}}_\phi = (-\sin\phi,\ \cos\phi,\ 0)\,. \tag{22}$$

The reported far-field components are

$$E_\theta^\infty = \widehat{\boldsymbol{e}}_\theta \cdot \boldsymbol{E}^\infty, \qquad E_\phi^\infty = \widehat{\boldsymbol{e}}_\phi \cdot \boldsymbol{E}^\infty. \tag{23}$$

The RCS uses the transverse magnitude

$$|\boldsymbol{E}^\infty|^2 = |E_\theta^\infty|^2 + |E_\phi^\infty|^2. \tag{24}$$

For the default $+z$-propagating incident wave, $\theta = 0°$ is the forward direction and $\theta = 180°$ is the backscatter direction. The default RCS comparison uses the E-plane cut

$$\phi = 0°, \qquad 0° \le \theta \le 180°.$$

Some papers use a different naming convention for the angular variables. If a paper says "$\theta = 0$, $0 \le \phi \le 180°$" for a plot that varies from forward to backscatter, its ϕ is the physical scattering angle. In DiscoGMPI, the same physical plot is produced with

$$\phi = 0°, \qquad \theta \in [0°, 180°].$$

At $\theta = 0°$ and $\theta = 180°$, the azimuthal basis is geometrically degenerate: all values of ϕ describe the same Cartesian observation direction, but the transverse basis rotates. A $\theta = 0°$, $0 \le \phi \le 180°$ plot is therefore a basis orientation check, not the usual forward-to-backward RCS cut.

6 Near-To-Far Transformation

6.1 Outgoing Green Phase

With the phasor convention $\mathrm{e}^{-\mathrm{i}\omega t}$, the outgoing scalar Green function has the phase

$$G(\boldsymbol{x},\boldsymbol{y}) = \frac{\mathrm{e}^{\mathrm{i}k|\boldsymbol{x}-\boldsymbol{y}|}}{4\pi|\boldsymbol{x}-\boldsymbol{y}|}. \tag{25}$$

For a far-field observation point $\boldsymbol{x} = r\widehat{\boldsymbol{r}}$,

$$|\boldsymbol{x}-\boldsymbol{y}| = r - \widehat{\boldsymbol{r}}\cdot\boldsymbol{y} + \mathcal{O}(r^{-1}). \tag{26}$$

Thus

$$\mathrm{e}^{\mathrm{i}k|\boldsymbol{x}-\boldsymbol{y}|} = \mathrm{e}^{\mathrm{i}kr}\mathrm{e}^{-\mathrm{i}k\widehat{\boldsymbol{r}}\cdot\boldsymbol{y}}\left(1+\mathcal{O}(r^{-1})\right). \tag{27}$$

This is why the DiscoGMPI near-to-far surface integral uses

$$\exp(-\mathrm{i}k\,\widehat{\boldsymbol{r}}\cdot\boldsymbol{y}) \tag{28}$$

for the spatial phase factor. Using the opposite sign would evaluate the same Fourier transform in direction $-\widehat{\boldsymbol{r}}$, effectively reflecting an E-plane RCS curve by the mapping

$$\theta \mapsto 180^\circ - \theta.$$

6.2 PEC Equivalent Surface Current

For a PEC scatterer, the relevant equivalent electric surface current is

$$\widehat{\boldsymbol{J}_s} = \boldsymbol{n}\times\widehat{\boldsymbol{H}_{\mathrm{tot}}} \qquad \text{on } \Gamma_{\mathrm{PEC}}. \tag{29}$$

The code evaluates this using the total magnetic field

$$\boldsymbol{H}_{\mathrm{tot},h}(\boldsymbol{x},t) = \boldsymbol{H}_{\mathrm{scat},h}(\boldsymbol{x},t) + \boldsymbol{H}_{\mathrm{inc}}(\boldsymbol{x},t). \tag{30}$$

The time-dependent surface-current integral for an observation direction $\widehat{\boldsymbol{r}}$ is

$$\boldsymbol{I}(\widehat{\boldsymbol{r}},t) = \int_{\Gamma_{\mathrm{PEC}}} \boldsymbol{J}_s(\boldsymbol{y},t)\mathrm{e}^{-\mathrm{i}k\widehat{\boldsymbol{r}}\cdot\boldsymbol{y}}\,\mathrm{d}S_y. \tag{31}$$

After Fourier extraction, the phasor integral is denoted $\widehat{\boldsymbol{I}}(\widehat{\boldsymbol{r}})$.

6.3 Far-Field Reconstruction

The far-field amplitude used in DiscoGMPI is

$$\boldsymbol{E}^{\infty}(\widehat{\boldsymbol{r}}) = \frac{\mathrm{i}kZ}{4\pi}\left[\widehat{\boldsymbol{r}}\left(\widehat{\boldsymbol{r}}\cdot\widehat{\boldsymbol{I}}(\widehat{\boldsymbol{r}})\right) - \widehat{\boldsymbol{I}}(\widehat{\boldsymbol{r}})\right]. \tag{32}$$

Equivalently,

$$\boldsymbol{E}^{\infty}(\widehat{\boldsymbol{r}}) = \frac{\mathrm{i}kZ}{4\pi}\widehat{\boldsymbol{r}}\times\left(\widehat{\boldsymbol{r}}\times\widehat{\boldsymbol{I}}(\widehat{\boldsymbol{r}})\right). \tag{33}$$

The vector in brackets is transverse because

$$\widehat{\boldsymbol{r}}\cdot\boldsymbol{E}^{\infty} = 0.$$

The overall sign convention depends on the normal convention and equivalent current convention. The RCS depends on $|\boldsymbol{E}^{\infty}|^2$, so a global sign does not affect the reported RCS.

Combining (2) and (32), the code reports

$$\sigma_h(\theta,\phi) = 4\pi\frac{|E^{\infty}_{\theta,h}|^2 + |E^{\infty}_{\phi,h}|^2}{A^2}. \tag{34}$$

7 Time-Domain Fourier Extraction

The solver advances real-valued fields in time. To obtain a monochromatic RCS, DiscoGMPI extracts the phasor at the incident frequency. If

$$f(t) = \mathrm{Re}\{\widehat{f}\,\mathrm{e}^{-\mathrm{i}\omega t}\},$$

then, over an integer number of periods,

$$\widehat{f} = \frac{2}{T}\int_0^T f(t)\,\mathrm{e}^{\mathrm{i}\omega t}\,\mathrm{d}t.$$

DiscoGMPI uses the discrete Hann-windowed estimator

$$\widehat{\boldsymbol{I}}_h(\widehat{\boldsymbol{r}}) = \frac{2}{W}\sum_m w_m\,\boldsymbol{I}_h(\widehat{\boldsymbol{r}}, t_m)\mathrm{e}^{\mathrm{i}\omega t_m}, \qquad W = \sum_m w_m. \tag{35}$$

The samples t_m are selected every `-rcs-every` time steps, and the final step is included. The current production drivers use a fixed adjusted time step, so the constant time-step factor cancels between numerator and denominator in (35).

The Hann window is

$$w(t) = \frac{1}{2}\left[1 - \cos\left(2\pi\frac{t - t_0}{T - t_0}\right)\right], \qquad t_0 \le t \le T, \tag{36}$$

where

$$t_0 = \texttt{-rcs-start-time}, \qquad T = \texttt{-final-time}.$$

Samples before t_0 are ignored. The purpose of t_0 is to exclude:

1. startup transients produced by abruptly imposing a time-harmonic incident field,
2. early nonperiodic adjustment of the scattered field,
3. wave packets reflected by the PML before they have decayed.

For a quantitative RCS comparison, the RCS curve should be stable when `-rcs-start-time` is shifted moderately later and when `-rcs-every` is reduced.

8 DG Surface Quadrature in the Code

The RCS calculation uses the same distributed DG mesh and polynomial basis as the time integration. The core data structure is `RCSFaceSample`, built by `build_pec_sphere_face_samples`. For each owned boundary face tagged as the metallic sphere:

1. The code keeps only faces with boundary tag `METALLIC_SPHERE_PEC_BOUNDARY_ID` $= 10$.
2. The face must belong to an element owned by the local MPI rank.
3. The face interpolation nodes are copied from the DG trace map.
4. The face quadrature weights are obtained by applying the reference face mass matrix to the constant vector 1, then scaling by physical face area:

$$w_q^{(F)} = \frac{|F|}{|F_{\mathrm{ref}}|}\,(M_F\mathbf{1})_q\,. \tag{37}$$

5. The physical coordinates of each face node are reconstructed by mapping the reference tetrahedron coordinates to the physical tetrahedron.

6. The inner-boundary DG normal is flipped to obtain the PEC scatterer normal used in $\boldsymbol{J}_s = \boldsymbol{n} \times \boldsymbol{H}_{\mathrm{tot}}$.

At one time sample t_m, the local rank contribution is

$$\boldsymbol{I}_{h,\mathrm{rank}}(\widehat{\boldsymbol{r}}, t_m) = \sum_{F \in \mathcal{F}_{\mathrm{rank}}^{\mathrm{PEC}}} \sum_{q \in F} w_q^{(F)} \left[\boldsymbol{n}_F \times \left(\boldsymbol{H}_{\mathrm{scat},h}(\boldsymbol{x}_{Fq}, t_m) + \boldsymbol{H}_{\mathrm{inc}}(\boldsymbol{x}_{Fq}, t_m)\right)\right] \mathrm{e}^{-\mathrm{i}k\widehat{\boldsymbol{r}}\cdot\boldsymbol{x}_{Fq}}. \tag{38}$$

The code accumulates this for every observation direction requested by the CLI. After multiplication by the Hann window and temporal Fourier phase, it adds the result to a rank-local complex accumulator:

$$\boldsymbol{A}_{\mathrm{rank}}(\widehat{\boldsymbol{r}}) \leftarrow \boldsymbol{A}_{\mathrm{rank}}(\widehat{\boldsymbol{r}}) + w_m \mathrm{e}^{\mathrm{i}\omega t_m} \boldsymbol{I}_{h,\mathrm{rank}}(\widehat{\boldsymbol{r}}, t_m). \tag{39}$$

The global accumulator is the MPI sum

$$\boldsymbol{A}(\widehat{\boldsymbol{r}}) = \sum_{\mathrm{rank}} \boldsymbol{A}_{\mathrm{rank}}(\widehat{\boldsymbol{r}}). \tag{40}$$

Finally,

$$\widehat{\boldsymbol{I}}_h(\widehat{\boldsymbol{r}}) = \frac{2}{W} \boldsymbol{A}(\widehat{\boldsymbol{r}}). \tag{41}$$

9 Algorithm Implemented in DiscoGMPI

The RCS path in `distributed_metallic_sphere_scattering.jl` can be summarized as follows.

1. Parse the scattering configuration: mesh, sphere radius, wavelength, frequency, amplitude, DG order, ESPRK order, CFL, PML parameters, flux choice, RCS angles, RCS sampling interval, and output directory.
2. Build the incident wave parameters:

$$\boldsymbol{d} = \widehat{\boldsymbol{z}}, \qquad \boldsymbol{p} = \widehat{\boldsymbol{x}}, \qquad k = 2\pi/\lambda, \qquad \omega = 2\pi f.$$

3. Load and partition the tetrahedral box-minus-sphere mesh.
4. Build the DG discretization and the incident-aware PEC boundary data.
5. Initialize the numerical unknown as the scattered field.
6. If RCS is enabled, build:
   - observation directions $\widehat{\boldsymbol{r}}(\theta, \phi)$,
   - transverse bases $\widehat{\boldsymbol{e}}_\theta$, $\widehat{\boldsymbol{e}}_\phi$,
   - owned PEC face samples,
   - complex current accumulators.
7. March the scattered field in time with the selected Poisson-bracket Maxwell/ESPRK path and PML source terms.
8. At every RCS sample time:
   - set the boundary-data time consistently,
   - reconstruct $\boldsymbol{H}_{\mathrm{tot},h} = \boldsymbol{H}_{\mathrm{scat},h} + \boldsymbol{H}_{\mathrm{inc}}$ on the PEC face nodes,

- compute $\boldsymbol{J}_{s,h} = \boldsymbol{n} \times \boldsymbol{H}_{\mathrm{tot},h}$,
- integrate $\boldsymbol{J}_{s,h} \mathrm{e}^{-\mathrm{i}k\widehat{\boldsymbol{r}}\cdot\boldsymbol{x}}$ over the owned PEC faces,
- multiply by $w_m \mathrm{e}^{\mathrm{i}\omega t_m}$,
- add to the complex accumulator.

9. At final time, reduce the complex accumulators across MPI ranks.
10. Reconstruct $E^{\infty}_{\theta,h}$, $E^{\infty}_{\phi,h}$, and σ_h for every observation angle.
11. Write `diagnostics/rcs.csv`.

10 Observation Grid and CLI Controls

The production driver controls the RCS observation directions with:

`-rcs-theta-count`
: Number of polar angle samples between `-rcs-theta-min-degrees` and `-rcs-theta-max-degrees`.

`-rcs-theta-min-degrees, -rcs-theta-max-degrees`
: Minimum and maximum polar scattering angle in degrees.

`-rcs-phi-degrees`
: Comma-separated list of azimuthal angles in degrees.

`-rcs-every`
: Time-step interval between RCS surface integrations.

`-rcs-start-time`
: First physical time included in the Fourier extraction.

For a smooth E-plane RCS plot comparable to the common PEC-sphere benchmark, one should use, for example,

```
--rcs-theta-count=181 \
--rcs-theta-min-degrees=0 \
--rcs-theta-max-degrees=180 \
--rcs-phi-degrees=0
```

This gives one-degree angular spacing. Increasing `-rcs-theta-count` makes the RCS plot smoother but increases the RCS surface-integration cost approximately linearly. It does not change the DG time integration itself.

11 Output Files

The main RCS diagnostic file is

`<run-dir>/diagnostics/rcs.csv`.

It contains one row per observation direction. The key columns are:

Column	Meaning
`theta_degrees`, `phi_degrees`	Observation angles.
`dir_x`, `dir_y`, `dir_z`	Cartesian components of $\widehat{\boldsymbol{r}}$.
`abs_E_infinity`	$\lvert\boldsymbol{E}_h^{\infty}\rvert$.
`abs_Etheta`, `abs_Ephi`	$\lvert E_{\theta,h}^{\infty}\rvert$, $\lvert E_{\phi,h}^{\infty}\rvert$.
`rcs`	σ_h.
`rcs_db`	$10\log_{10}(\sigma_h)$, not normalized by πa^2.
`real_Etheta`, `imag_Etheta`	Complex $E_{\theta,h}^{\infty}$.
`real_Ephi`, `imag_Ephi`	Complex $E_{\phi,h}^{\infty}$.
`samples`	Number of nonzero-window RCS time samples.
`window_sum`	$W = \sum_m w_m$.
`rcs_start_time`, `rcs_end_time`	Fourier window endpoints.

The Python postprocessor writes:

- `diagnostics/mie_rcs_comparison.csv`,
- `tables/mie_rcs_comparison.tex`,
- `plots/mie_rcs_comparison.pdf`,
- `plots/mie_rcs_polar.pdf`,
- `plots/mie_rcs_theta0_phi_cut.pdf`.

The normalized plotted quantity is

$$10\log_{10}\left(\frac{\sigma}{\pi a^2}\right),$$

even though the raw `rcs_db` column in `rcs.csv` is the dB value of σ itself.

For the NASA PEC Almond workflow, `examples/postprocess_pec_almond_rcs.py` writes analogous comparison artifacts using the Austin normalization σ/λ_0^2, not $\sigma/(\pi a^2)$. The relevant outputs are:

- `diagnostics/pec_almond_rcs_comparison.csv`,
- LaTeX comparison tables under `tables/`,
- Cartesian comparison plots under `plots/`,
- optional PNG versions when `-write-png` is provided.

The almond comparison can be made against Austin simulation data `ref`, direct measurement data `meas-m1`, or scaled measurement data `meas-m2`.

12 Example RCS Figures

Figures 2 and 3 show the postprocessed output from

`output/validation_sequence/coarse_sphere_rcs_epw2_p4_transient_theta181/plots`.

The file `mie_rcs_theta0_phi_cut.pdf` is intentionally not included here because, at $\theta = 0°$, varying ϕ rotates the transverse basis but does not change the Cartesian observation direction. The two plots included below are the physically relevant forward-to-backward E-plane RCS views for the default $\phi = 0°$ cut.

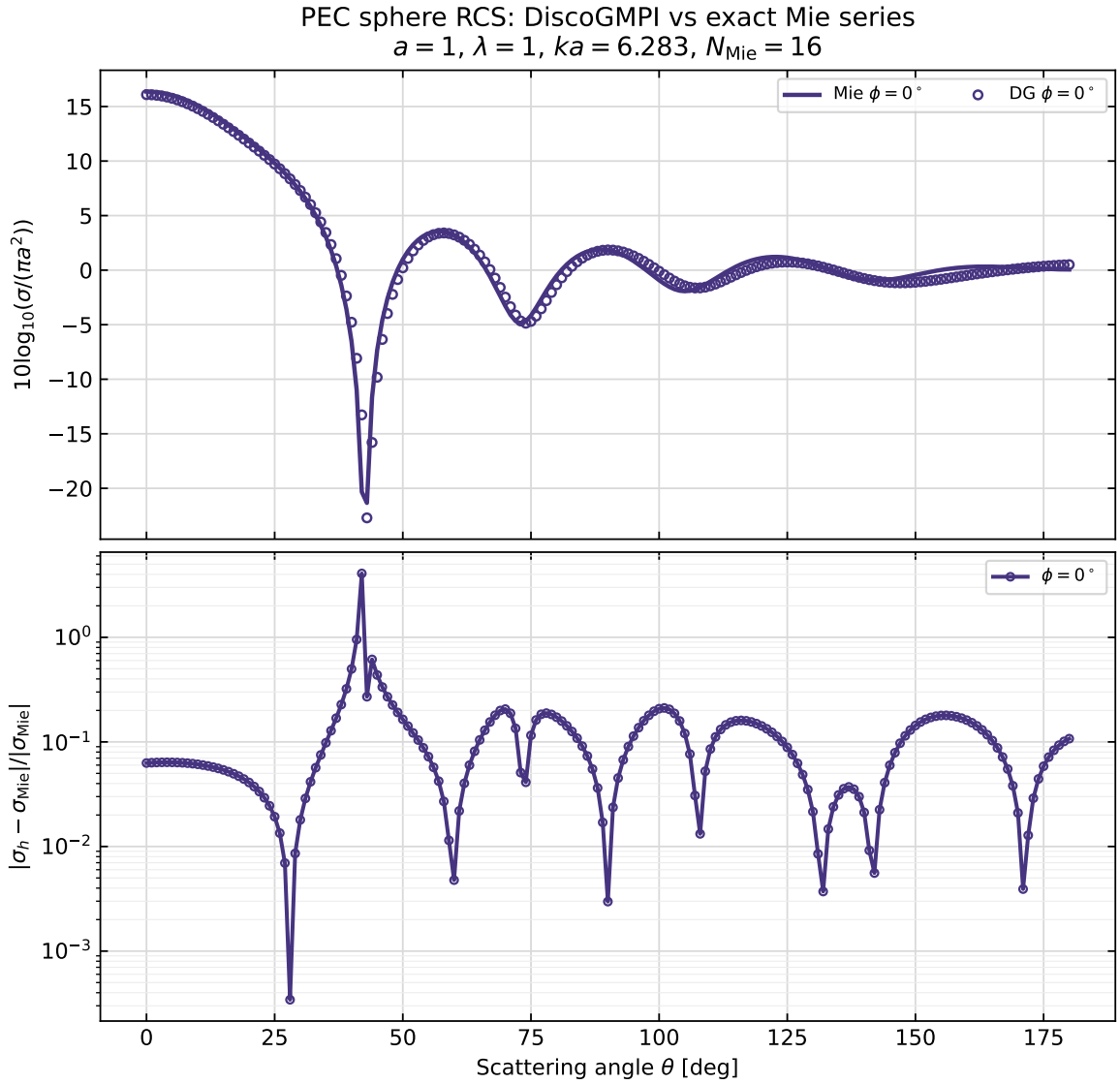


Figure 2: Cartesian RCS comparison for the `coarse_sphere_rcs_epw2_p4_transient_theta181` run. The upper panel compares $10\log_{10}(\sigma/(\pi a^2))$ for the DG near-to-far extraction and the exact PEC-sphere Mie series. The lower panel reports the relative RCS error.

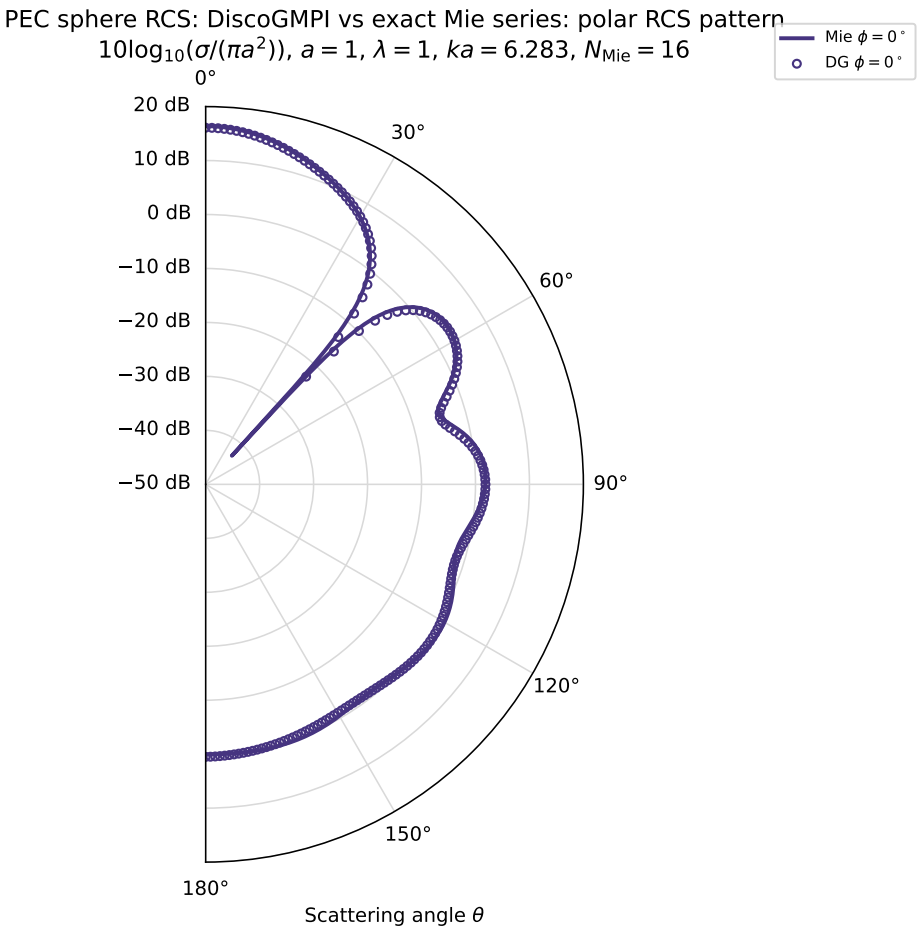


Figure 3: Polar RCS pattern for the same run. The angular coordinate is the DiscoGMPI polar scattering angle θ, with $\theta = 0^\circ$ denoting the forward $+z$ direction and $\theta = 180^\circ$ denoting backscatter. The radial tick labels are normalized dB values, $10\log_{10}(\sigma/(\pi a^2))$.

13 MN5 EPW4/P4 RCS Validation Run

The directory

`output/MN5/coarse_sphere_rcs_epw4_p4_transient_theta181`

contains an eight-rank MN5 execution of the metallic-sphere workflow. This is the higher-resolution run currently used as the main DiscoGMPI RCS validation case. The computation solves for the scattered field around a PEC sphere, adds the analytical incident wave for output and boundary data, extracts the time-harmonic scattered far field after the transient, and compares the result against the exact PEC-sphere Mie series.

Quantity	Value
Driver	`distributed_metallic_sphere_scattering` through the coarse RCS validation wrapper.
Mesh	`examples/meshes/metallic_sphere_scattering_epw4.vtk`.
Geometry	PEC sphere of radius $a = 1$ centered in the box $[-3, 3]^3$.
Absorbing layer	Cartesian nonlinear PML on the six outer box slabs, with width 1, $\sigma_{\max} = 12$, polynomial degree 2, and $a_{\mathrm{PML}} = 0.5$.
Incident wave	$+z$-propagating, x-polarized plane wave with amplitude 1, wavelength $\lambda = 1$, frequency $f = 1$, and $\epsilon = \mu = 1$.
Discretization	DG order $p = 4$, 35 nodes per tetrahedron, centered flux, ESPRK order 5, CFL 0.15.
Mesh size and time step	15673 global tetrahedra, $h_{\min} = 2.0491638407152006 \times 10^{-2}$, $\Delta t = 7.683934174297241 \times 10^{-4}$.
Time interval	$T = 12$, 15617 time steps.
MPI layout	8 ranks, owned-element imbalance 1.00051, with 1959–1960 owned elements per rank.
RCS extraction	Fourier window starts at $t = 5$, samples every 5 time steps, $\theta = 0^\circ, \ldots, 180^\circ$ with 181 angles, and $\phi = 0^\circ$.
Runtime	16477.35 s, approximately 4.58 hours.

The Mie comparison summary reports a successful validation gate for the available criteria: all 181 angular samples are finite and positive, the largest numerical RCS is 1.33865×10^2, and the Fourier window contains 1822 samples with a window sum of 910.992. For the PEC sphere with $ka = 2\pi$, the postprocessor used 16 Mie terms. The RMS relative error over the $\phi = 0^\circ$ forward-to-backward cut is 6.95×10^{-2}, the maximum relative error is 5.97×10^{-1}, the maximum dB error is 3.16 dB, and the maximum absolute error normalized by πa^2 is 7.25×10^{-1}. The largest relative errors occur near deep Mie nulls, where the exact RCS is small and relative normalization amplifies small absolute differences.

The same run also wrote time-series diagnostics and field visualization data. The final scattered energy was 7.0974, and the final PEC boundary residual diagnostic gave a relative RMS value 1.90×10^{-1}. This residual is a useful boundary-consistency diagnostic, but the RCS validation is primarily assessed by the far-field comparison against the Mie reference after the transient window begins.

14 NASA/Austin PEC Almond Benchmark

The NASA PEC Almond case in DiscoGMPI targets Problem Set IIIA of the Austin RCS Benchmark Suite. The object is a nontrivial perfectly electrically conducting almond with a sharp tip and a slender body. This geometry is much more demanding than the PEC sphere because there is no analytic Mie reference, the surface curvature varies strongly, and the tip can create very small tetrahedra if the imported surface mesh is not controlled.

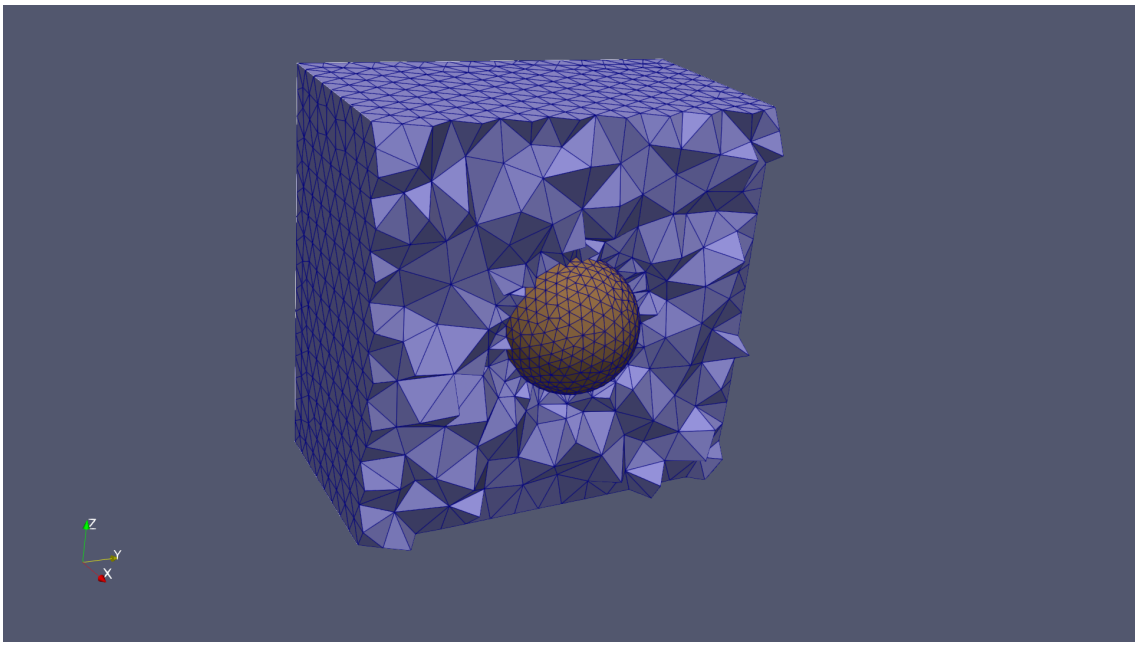


Figure 4: Mesh visualization for the MN5 EPW4/P4 PEC-sphere RCS run. The picture shows the tetrahedral box-minus-sphere discretization used by `examples/meshes/metallic_sphere_scattering_epw4.vtk`. The inner spherical boundary is the PEC scatterer on which the incident-aware boundary trace is imposed and from which the surface-current RCS integral is sampled. The surrounding Cartesian box provides the outer PML slabs. This figure is mainly a geometry and mesh-quality check: the Mie comparison is only meaningful if the spherical PEC surface and the near-target volume mesh are resolved consistently with the requested EPW4 discretization.

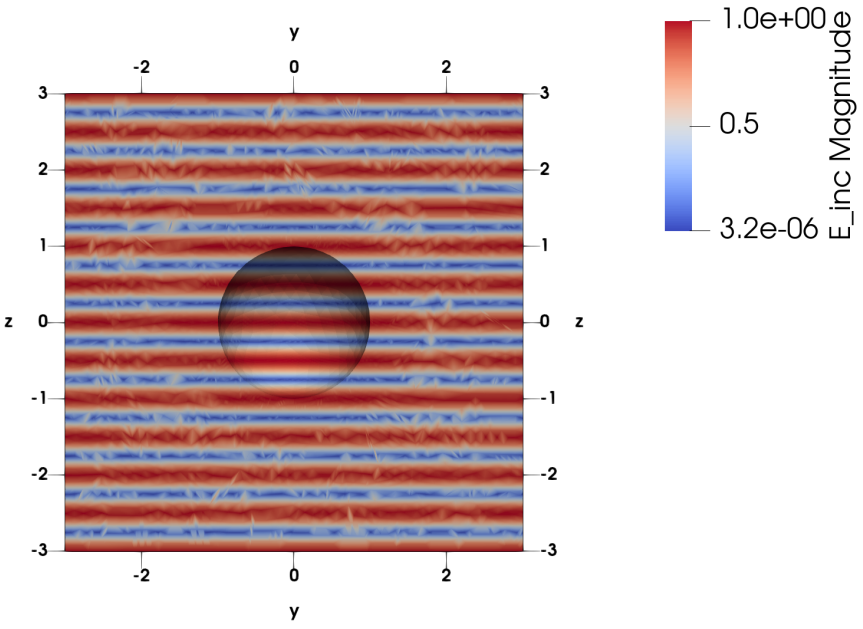


Figure 5: Incident-field visualization from the MN5 EPW4/P4 run. The plot shows the imposed $+z$-propagating, x-polarized plane wave used to construct the total field and the incident-aware PEC boundary condition.

14.1 Benchmark Definition

The benchmark object is a PEC almond proportional to the NASA benchmark almond dimensions. The smallest benchmark size used by the available reference data is

$$L = 9.936\,\text{in} = 0.2523744\,\text{m}. \tag{42}$$

The Austin data available in this repository cover the four frequency keys

Frequency key	Frequency	Free-space wavelength
`fx1`	3.5 GHz	8.5654988×10^{-2} m
`fx2`	5.125 GHz	5.8496089×10^{-2} m
`fx3`	7.0 GHz	4.2827494×10^{-2} m
`fx4`	10.25 GHz	2.9248045×10^{-2} m

The default DiscoGMPI almond workflow uses `fx1`, so

$$\lambda_0 = \frac{c_0}{3.5 \times 10^9} = 8.5654988 \times 10^{-2}\,\text{m}.$$

The Austin benchmark quantity of interest is monostatic RCS on the $\theta = 90^\circ$ aspect cut, with

$$0^\circ \le \phi \le 180^\circ.$$

The reference simulation files are sampled every 0.5°, giving 361 angular samples. The provided measurement files are sampled more densely, every 0.25°. The data-file columns are

$$\text{frequency (Hz)}, \quad \theta\ \text{(deg)}, \quad \phi\ \text{(deg)}, \quad \sigma_{\text{dBsm}}. \tag{43}$$

The reference files live in

`benchmark/AustinCEMBenchmarks/Problem_III-Almonds/IIIA-ReferenceData`.

Their naming convention is:

- `ref_rcs.III.A.s1.fxK.H.txt` and `ref_rcs.III.A.s1.fxK.V.txt`: reference simulation data;
- `meas.m1_rcs.III.A.s1.fxK.H.txt` and `meas.m1_rcs.III.A.s1.fxK.V.txt`: measurements of the $L = 9.936$ in almond;
- `meas.m2_rcs.III.A.s1.fxK.H.txt` and `meas.m2_rcs.III.A.s1.fxK.V.txt`: measurements of a two-times larger almond shifted by $10\log_{10} 4$ dB and by the corresponding frequency scaling.

Here `K` $\in \{1, 2, 3, 4\}$. The suffix `H` denotes HH polarization, and the suffix `V` denotes VV polarization.

14.2 Polarization and Normalization

The Austin notation uses $\sigma_{\phi\phi}$ for HH polarization and $\sigma_{\theta\theta}$ for VV polarization. The DiscoGMPI almond postprocessor maps these to the far-field components

$$\text{H/HH} \longleftrightarrow E_\phi^\infty, \qquad \text{V/VV} \longleftrightarrow E_\theta^\infty. \tag{44}$$

The raw DiscoGMPI RCS file stores total RCS and the component far-field magnitudes. The almond postprocessor reconstructs component RCS values using

$$\sigma_{\theta,h} = 4\pi \frac{|E_{\theta,h}^\infty|^2}{A^2}, \tag{45}$$

$$\sigma_{\phi,h} = 4\pi \frac{|E_{\phi,h}^\infty|^2}{A^2}. \tag{46}$$

The comparison plots use the dimensionless benchmark-style normalization

$$10 \log_{10} \left(\frac{\sigma}{\lambda_0^2} \right), \tag{47}$$

while the Austin text files store σ directly in dBsm. Therefore the postprocessor converts a reference dBsm value by

$$\sigma_{\lambda,\mathrm{dB}} = \sigma_{\mathrm{dBsm}} - 20 \log_{10}(\lambda_0). \tag{48}$$

14.3 DiscoGMPI Almond Workflow

The production frontend is

`examples/distributed_pec_almond_scattering.jl`.

It is intentionally thin: it reuses the generic PEC-scatterer Maxwell/PML/RCS driver in `examples/distributed_metallic_sphere_scattering.jl`. The almond frontend supplies:

- the default almond mesh, `examples/meshes/pec_almond_scattering_fx1_epw4.vtk`;
- the PEC boundary tag 10;
- the scatterer name `PEC almond`;
- radius 0, which disables the sphere-specific radius check;
- a physical frequency-to-wavelength conversion $\lambda_0 = c_0/f$;
- one-wavelength Cartesian PML width by default;
- RCS extraction enabled by default.

The solver itself remains nondimensional. With $\epsilon = \mu = 1$, the internal wave speed is 1. The physical benchmark frequency is used to set the numerical wavelength; for example,

$$f = 3.5\,\mathrm{GHz} \quad \Rightarrow \quad \lambda_0 = 8.5654988 \times 10^{-2}.$$

The normalized solver angular frequency is then $\omega = 2\pi/\lambda_0$.

The current Austin-style SLURM workflow is

`examples/run_nasa_almond_rcs.slurm`.

Its defaults run

- `fx1`: 3.5 GHz;
- DG order $p = 2$, ESPRK order 3;
- centered Poisson-bracket flux;
- final time $12\lambda_0$;
- RCS Fourier window start time $5\lambda_0$;
- $\theta = 90°$, $\phi = 0°, 0.5°, \ldots, 180°$;
- RCS sampling every 5 time steps;
- diagnostics every 250 time steps;
- ParaView output disabled unless `PARAVIEW_EVERY` is overridden.

The postprocessor is

`examples/postprocess_pec_almond_rcs.py`.

It reads `diagnostics/rcs.csv`, selects the requested angular cut, compares against Austin `ref`, `meas-m1`, or `meas-m2` files, and writes CSV, LaTeX, PNG, and PDF outputs in the run directory.

14.4 Current Default Mesh

The mesh generator is

`examples/meshes/generate_pec_almond_scattering_mesh.sh`

with geometry body

`examples/meshes/pec_almond_scattering.geo`.

The geometry is a Cartesian box with an inner PEC almond cavity. The six outer faces are tagged as PML boundaries and the almond surface is tagged as PEC:

Physical tag	Meaning
$1, \ldots, 6$	outer Cartesian box faces, used by the Cartesian PML;
10	inner PEC almond boundary;
7	air plus PML volume.

The current default EPW4 mesh `examples/meshes/pec_almond_scattering_fx1_epw4.vtk` has:

Quantity	Value
Points	9937
VTK cells	50331
Boundary triangles	16648
PEC almond triangles, tag 10	14300
Outer PML-boundary triangles, tags $1, \ldots, 6$	2348
Auxiliary vertex/line cells, tag -1	166
Tetrahedra, tag 7	33517
Domain lower corner	$(-0.297497176, -0.220090139, -0.187570926)$
Domain upper corner	$(0.297497176, 0.220090139, 0.187570926)$
Minimum tet edge	2.479743×10^{-5}
First-percentile tet edge	3.553260×10^{-4}
Median tet edge	1.170343×10^{-2}
$\min(3V/A)$	1.505260×10^{-6}
First-percentile $3V/A$	3.088514×10^{-5}
Median $3V/A$	6.517538×10^{-4}

Here $3V/A$ is the element-size measure used by `estimate_maxwell_dt`; it is the actual explicit-timestep limiter in the current driver.

14.4.1 Benchmark Mesh Figures

The Austin benchmark folder contains four PNG views of the NASA almond mesh. They are included in Figures 10, 11, 12, and 13 because they document the geometric features that control both accuracy and runtime:

- the global volume-mesh view shows the tetrahedral volume around the PEC cavity and the contrast between near-target and far-field mesh sizes;
- the xy-plane view shows the planform almond shape and the sharp downstream tip, which is the main source of small elements;
- the xz-plane view shows the slender longitudinal profile and the thin vertical thickness;
- the yz-plane view shows the cross-section, useful for checking whether the mesh resolves the transverse thickness without producing pathological slivers.

14.5 Reference Comparison and Error Metrics

Unlike the PEC sphere, the almond has no closed-form Mie-series reference. The quantitative reference is the benchmark data set. The DiscoGMPI postprocessor computes a row-wise dB error

$$e_{\mathrm{dB}}(\phi_j) = \sigma_{h,\lambda,\mathrm{dB}}(\phi_j) - \sigma_{\mathrm{ref},\lambda,\mathrm{dB}}(\phi_j). \tag{49}$$

The Austin benchmark also defines a thresholded dB error that discounts differences in regions where the reference RCS is more than 80 dB below the maximum reference level. For polarization $u \in \{\theta, \phi\}$, let

$$T_{u,\mathrm{dB}} = \max_{\phi} \sigma^{\mathrm{ref}}_{uu,\mathrm{dB}}(\phi) - 80. \tag{50}$$

Then the thresholded curves are

$$\sigma^{\mathrm{TH}}_{uu,\mathrm{dB}}(\phi) = \max\left(\sigma_{uu,\mathrm{dB}}(\phi), T_{u,\mathrm{dB}}\right), \tag{51}$$

and a benchmark-style average error is

$$\overline{e}^{\mathrm{TH}}_{uu,\mathrm{dB}} = \frac{1}{N_\phi} \sum_{j=1}^{N_\phi} \left|\sigma^{\mathrm{TH}}_{h,uu,\mathrm{dB}}(\phi_j) - \sigma^{\mathrm{TH}}_{\mathrm{ref},uu,\mathrm{dB}}(\phi_j)\right|. \tag{52}$$

This thresholding is important because deep RCS nulls can dominate relative or dB errors even when the absolute scattered field is small.

14.6 Interpretation and Current Limitations

The almond workflow should be interpreted differently from the sphere workflow:

- The sphere comparison is an exact-solution validation; the almond comparison is a benchmark-data validation.
- The sphere is geometrically smooth; the almond has sharp features, so mesh quality and the minimum $3V/A$ element size strongly affect runtime.
- The Austin data are monostatic aspect data. The current DiscoGMPI script samples the implemented RCS diagnostic on the $\theta = 90°$, $0 \le \phi \le 180°$ cut and compares the corresponding E^{∞}_{ϕ} or E^{∞}_{θ} component to the selected Austin reference. A strict monostatic aspect sweep requires the incident direction and observation direction to be swept consistently, or equivalently the target to be rotated against a fixed incident wave.
- Quantitative almond validation should therefore be reported with the exact run convention, mesh, frequency key, polarization, reference type, RCS window, and angular sampling.

14.7 Recommended Almond Commands

Generate the default almond mesh with:

```
bash examples/meshes/generate_pec_almond_scattering_mesh.sh
```

Run the default Austin-style RCS workflow on a SLURM system with:

```
sbatch examples/run_nasa_almond_rcs.slurm
```

Useful overrides are:

```
sbatch --export=ALL,ORDER=4,ESPRK_ORDER=5,FINAL_TIME=1.027859856,\
RCS_START_TIME=0.42827494,PARAVIEW_EVERY=500 \
  examples/run_nasa_almond_rcs.slurm
```

For an interactive MPI run without the SLURM wrapper:

```
mpiexec -n 8 -genv OPENBLAS_NUM_THREADS 1 julia --project=. \
  examples/distributed_pec_almond_scattering.jl \
  --mesh=examples/meshes/pec_almond_scattering_fx1_epw4.vtk \
  --frequency-hz=3.5e9 \
  --order=2 --esprk-order=3 \
  --final-time=1.027859856 \
  --rcs-start-time=0.42827494 \
  --rcs-every=5 \
  --rcs-theta-count=1 \
  --rcs-theta-min-degrees=90 \
  --rcs-theta-max-degrees=90 \
  --rcs-phi-degrees="$(python3 -c 'print(",".join(f"{0.5*i:.1f}" for i in range(361)))
    ')" \
  --diagnostics-every=250 \
  --paraview-every=0 \
  --output-dir=output/validation_sequence/pec_almond_rcs_fx1_p2_theta90_phi0_180
```

Then compare against both HH and VV Austin reference curves:

```
python3 examples/postprocess_pec_almond_rcs.py \
  --run-dir=output/validation_sequence/pec_almond_rcs_fx1_p2_theta90_phi0_180 \
  --frequency-hz=3.5e9 \
  --frequency-key=fx1 \
  --reference-kind=ref \
  --polarization=both \
  --dg-angle=phi \
  --theta-filter=90 \
  --write-png
```

15 Exact PEC-Sphere Mie Reference

For a PEC sphere, the exact frequency-domain solution is given by the Mie series. The post-processor evaluates the scattering amplitude functions $S_1(\theta)$ and $S_2(\theta)$ using the PEC Mie coefficients. With size parameter

$$x = ka,$$

the scalar Riccati-Bessel functions are

$$\psi_n(x) = x j_n(x), \qquad \xi_n(x) = x h_n^{(1)}(x),$$

and the PEC coefficients used by the code are

$$a_n = -\frac{\psi_n'(x)}{\xi_n'(x)}, \qquad b_n = -\frac{\psi_n(x)}{\xi_n(x)}. \tag{53}$$

The angular functions are

$$\pi_1(\mu) = 1, \tag{54}$$

$$\pi_2(\mu) = 3\mu, \tag{55}$$

$$\pi_n(\mu) = \frac{2n-1}{n-1}\mu\,\pi_{n-1}(\mu) - \frac{n}{n-1}\pi_{n-2}(\mu), \qquad n \geq 3, \tag{56}$$

$$\tau_n(\mu) = n\mu\,\pi_n(\mu) - (n+1)\pi_{n-1}(\mu), \qquad \mu = \cos\theta. \tag{57}$$

The scattering amplitudes are

$$S_1(\theta) = \sum_{n=1}^{N} \frac{2n+1}{n(n+1)} \left[a_n \pi_n(\cos\theta) + b_n \tau_n(\cos\theta) \right], \tag{58}$$

$$S_2(\theta) = \sum_{n=1}^{N} \frac{2n+1}{n(n+1)} \left[a_n \tau_n(\cos\theta) + b_n \pi_n(\cos\theta) \right]. \tag{59}$$

For the DiscoGMPI default incident wave, the postprocessor maps these amplitudes to the same $(\widehat{\boldsymbol{e}}_\theta, \widehat{\boldsymbol{e}}_\phi)$ basis as the numerical RCS:

$$E^{\infty}_{\theta,\mathrm{Mie}} = \frac{S_2(\theta)\cos\phi}{k}, \qquad E^{\infty}_{\phi,\mathrm{Mie}} = -\frac{S_1(\theta)\sin\phi}{k}. \tag{60}$$

Then

$$\sigma_{\mathrm{Mie}}(\theta,\phi) = 4\pi \left(|E^{\infty}_{\theta,\mathrm{Mie}}|^2 + |E^{\infty}_{\phi,\mathrm{Mie}}|^2 \right). \tag{61}$$

The automatic truncation order is

$$N = \max\left(8,\ \left\lceil x + 4x^{1/3} + 2 \right\rceil\right). \tag{62}$$

16 Validation Strategy

A correct RCS workflow must pass several levels of validation.

16.1 Boundary Residual

Before trusting RCS data, verify the incident-aware PEC boundary condition:

$$R_{\mathrm{PEC}} = \left\| \boldsymbol{n} \times (\boldsymbol{E}_{\mathrm{scat},h} + \boldsymbol{E}_{\mathrm{inc}}) \right\|_{L^2(\Gamma_{\mathrm{PEC}})}. \tag{63}$$

The validation driver `examples/validate_pec_sphere_boundary_residual.jl` writes `diagnostics/pec_boun`
A large residual means the surface current is being evaluated on a field that does not satisfy the PEC condition accurately.

For the NASA PEC Almond, use `examples/validate_pec_almond_boundary_residual.jl`. It evaluates the same residual on the almond surface tag 10. The almond residual is especially useful because errors near the sharp tip can contaminate the surface current used for RCS extraction.

16.2 Qualitative Near Field

Inspect $\boldsymbol{E}_{\mathrm{scat}}$, $\boldsymbol{H}_{\mathrm{scat}}$, $\boldsymbol{E}_{\mathrm{tot}}$, and $\boldsymbol{H}_{\mathrm{tot}}$ in ParaView. The production driver can write:

- $\boldsymbol{E}_{\mathrm{scat}}, \boldsymbol{H}_{\mathrm{scat}}$,
- $\boldsymbol{E}_{\mathrm{inc}}, \boldsymbol{H}_{\mathrm{inc}}$,
- $\boldsymbol{E}_{\mathrm{tot}}, \boldsymbol{H}_{\mathrm{tot}}$,
- $|\boldsymbol{E}_{\mathrm{scat}}|, |\boldsymbol{H}_{\mathrm{scat}}|$.

For the default plane wave, the components E_x and H_y are the most direct comparison with the metallic-sphere benchmark figures.

16.3 Mie Comparison

After a run finishes, use

```
python3 examples/postprocess_metallic_sphere_mie_rcs.py \
  --run-dir <run-dir>
```

The comparison figure should be interpreted as:

- top panel: numerical and Mie $10\log_{10}(\sigma/(\pi a^2))$;
- bottom panel:

$$\frac{|\sigma_h - \sigma_{\mathrm{Mie}}|}{|\sigma_{\mathrm{Mie}}|}.$$

For quantitative validation, repeat the run with:

1. later `-rcs-start-time`,
2. smaller `-rcs-every`,
3. more angular samples,
4. finer sphere geometry,
5. larger polynomial order,
6. lower PML reflection.

A reliable RCS estimate should not change significantly under these variations.

16.4 Austin Almond Reference Comparison

For the almond, use:

```
python3 examples/postprocess_pec_almond_rcs.py \
  --run-dir <run-dir> \
  --frequency-key fx1 \
  --reference-kind ref \
  --polarization both \
  --dg-angle phi \
  --theta-filter 90
```

This comparison is not an exact-solution validation. It is a benchmark-data comparison against Austin simulation or measurement files. Report at least:

- frequency key and physical frequency;
- reference kind, `ref`, `meas-m1`, or `meas-m2`;
- polarization, H/HH or V/VV;
- angular cut and sampling;
- RCS window start time and number of Fourier samples;
- mesh name, DG order, and MPI rank count;
- mean and maximum dB error, preferably with the Austin thresholded metric when reporting benchmark performance.

17 Recommended Commands

A high-order EPW2/EPW4 smoke or development run can be launched with:

```
mpiexec -n 2 -genv OPENBLAS_NUM_THREADS 1 julia --project=. \
  examples/validate_coarse_sphere_rcs.jl \
  --mesh=examples/meshes/metallic_sphere_scattering_epw2.vtk \
  --order=4 --esprk-order=5 \
  --final-time=12.0 --rcs-start-time=5.0 \
  --rcs-every=5 --diagnostics-every=250 \
  --rcs-theta-count=181 \
  --rcs-theta-min-degrees=0 \
  --rcs-theta-max-degrees=180 \
  --rcs-phi-degrees=0 \
  --min-max-rcs=0 \
  --output-dir=output/validation_sequence/sphere_rcs_epw2_p4_theta181
```

Then postprocess:

```
python3 examples/postprocess_metallic_sphere_mie_rcs.py \
  --run-dir output/validation_sequence/sphere_rcs_epw2_p4_theta181
```

For a publication-quality RCS curve, EPW2 is usually too coarse geometrically. Use at least EPW3 or EPW4, enough runtime for a steady monochromatic response, and an RCS window that excludes startup transients.

For the NASA PEC Almond benchmark, prefer the SLURM workflow in `examples/run_nasa_almond_rcs.slurm`; it sets the Austin $\theta = 90°$, $0° \leq \phi \leq 180°$ angular cut and runs the almond postprocessor automatically. See Section 14 for the full command sequence and caveats.

18 Common Failure Modes

Too few points in the RCS curve.
Increase `-rcs-theta-count`. The postprocessor only plots the angles present in `diagnostics/rcs.csv`.

Mie and DG curves are mirrored left-to-right.
Check the near-to-far phase sign. For the convention $\cos(kz - \omega t)$, the surface integral uses $\exp(-\mathrm{i}k\widehat{\boldsymbol{r}} \cdot \boldsymbol{x})$.

Large error near deep Mie nulls.
Relative error is sensitive when σ_{Mie} is very small. Inspect absolute normalized error and dB error as well.

Good boundary residual but poor RCS.
Possible causes are insufficient sphere geometry resolution, PML reflection, poor Fourier window, too few periods in the accumulation window, or too coarse angular sampling.

Good early RCS but later contamination.
Reflections from the PML or outer boundary may be returning to the sphere. Increase PML quality, enlarge the buffer, or end the Fourier window before reflections return.

No RCS rows.
RCS was disabled, the run failed before final output, no faces were tagged with the metallic sphere boundary ID, or the Fourier window contained no nonzero samples.

Confusion between θ and ϕ.
In DiscoGMPI, θ is the polar scattering angle from $+z$, and ϕ is the azimuthal angle. The standard E-plane sphere plot is $\phi = 0$ with $\theta \in [0°, 180°]$.

Almond HH/VV curves appear swapped.

Check the polarization mapping. In the almond postprocessor, `H` means HH and uses E_ϕ^∞, while `V` means VV and uses E_θ^∞.

Almond comparison has no matching reference rows.

Check `-frequency-key`, `-reference-kind`, and the selected angular cut. The Austin reference simulation files use $\theta = 90^\circ$ and $\phi = 0^\circ, 0.5^\circ, \ldots, 180^\circ$.

Almond run is much slower than expected.

Inspect the mesh-size diagnostics, especially $\min(3V/A)$. A few sliver tetrahedra near the almond tip can reduce the explicit time step by orders of magnitude even when the average edge length looks reasonable.

Good almond curve shape but persistent dB offset.

Verify the normalization. The almond plots use σ/λ_0^2; the sphere plots use $\sigma/(\pi a^2)$. Mixing these normalizations creates a constant vertical shift in dB.

19 Checklist for a Quantitative RCS Study

1. Verify the mesh: PEC boundary tag 10, adequate target surface resolution, adequate PML/buffer region, and no pathological $\min(3V/A)$ outliers.
2. Verify the incident wave: direction $+z$, polarization x, wavelength, frequency, amplitude, ϵ, and μ.
3. Verify PEC residual: $\|\boldsymbol{n} \times \boldsymbol{E}_{\text{tot}}\|_{L^2(\Gamma_{\text{PEC}})}$.
4. Run long enough for a periodic scattered field.
5. Set `-rcs-start-time` after the startup transient.
6. Use enough angular samples, e.g. `-rcs-theta-count=181` or `361`.
7. Compare against the corrected Mie postprocessor.
8. Repeat with a later `-rcs-start-time`.
9. Repeat with a smaller `-rcs-every`.
10. Repeat on a finer mesh or with a higher-order discretization.
11. Confirm that the RCS curve and error metrics stabilize.
12. For the almond, verify the benchmark convention separately: $\theta = 90^\circ$, $\phi \in [0^\circ, 180^\circ]$, H/HH mapped to E_ϕ^∞, V/VV mapped to E_θ^∞, and σ/λ_0^2 normalization.

20 Code Map

The main implementation locations are:

File or function	Role
`examples/distributed_metallic_sphere_scattering.jl`	Production scattering and RCS driver.
`RCSObservationDirection`	Stores (θ, ϕ), $\widehat{\boldsymbol{r}}$, $\widehat{\boldsymbol{e}}_\theta$, $\widehat{\boldsymbol{e}}_\phi$.
`RCSFaceSample`	Stores owned PEC face nodes, weights, coordinates, and normals.
`build_pec_sphere_face_samples`	Extracts tagged sphere faces and flips the inner-boundary normal.
`accumulate_rcs_sample!`	Computes the windowed surface-current Fourier contribution.
`global_rcs_accumulator`	MPI reduction of complex accumulators.
`far_field_from_surface_current`	Applies (32).
`write_rcs_results`	Writes `diagnostics/rcs.csv`.
`src/MaxwellExactSolutions.jl`	Incident plane-wave definitions.
`examples/validate_coarse_sphere_rcs.jl`	Coarse validation gate and Julia Mie comparison.
`examples/postprocess_metallic_sphere_mie_rcs.py`	Publication-style Mie comparison plots and tables.
`examples/distributed_pec_almond_scattering.jl`	NASA/Austin PEC Almond frontend over the generic PEC-scatterer driver.
`examples/meshes/pec_almond_scattering.geo`	Box-minus-almond Gmsh geometry with PML outer tags and PEC almond tag.
`examples/meshes/generate_pec_almond_scattering_mesh.sh`	Reproducible almond mesh-generation wrapper.
`examples/validate_pec_almond_boundary_residual.jl`	Almond PEC residual gate.
`examples/validate_pec_almond_rcs_smoke.jl`	Short almond RCS smoke gate for finite, positive, angle-complete output.
`examples/postprocess_pec_almond_rcs.py`	Austin benchmark comparison for HH/VV almond RCS data.
`examples/run_nasa_almond_rcs.slurm`	End-to-end SLURM workflow: run almond RCS and postprocess against Austin data.
`benchmark/AustinCEMBenchmarks/Problem_III-Almonds`	Local Austin benchmark description, reference data, and mesh figures.

21 Summary

DiscoGMPI computes PEC-target RCS by solving the scattered-field Maxwell problem in the time domain, reconstructing the total magnetic field on the metallic surface, integrating the equivalent PEC surface current with the outgoing far-field phase $\exp(-\mathrm{i}k\widehat{\boldsymbol{r}} \cdot \boldsymbol{x})$, extracting the phasor at the incident frequency with a Hann-windowed Fourier average, and normalizing the far-field amplitude according to

$$\sigma = 4\pi |\boldsymbol{E}^\infty|^2 / A^2.$$

The procedure is sensitive to phase convention, angular convention, surface normal orientation, mesh geometry, PML reflections, and Fourier-window selection. The corrected Mie postprocessor provides the primary quantitative reference for the PEC sphere, but a trustworthy comparison requires a resolved mesh and a transient-free Fourier window. The NASA PEC Almond workflow uses the same near-to-far machinery, but validation is against Austin benchmark data rather than an exact solution; the important additional ingredients are the HH/VV component mapping, σ/λ_0^2 normalization, the $\theta = 90^\circ$ azimuthal cut, and careful control of the almond mesh near the sharp tip.

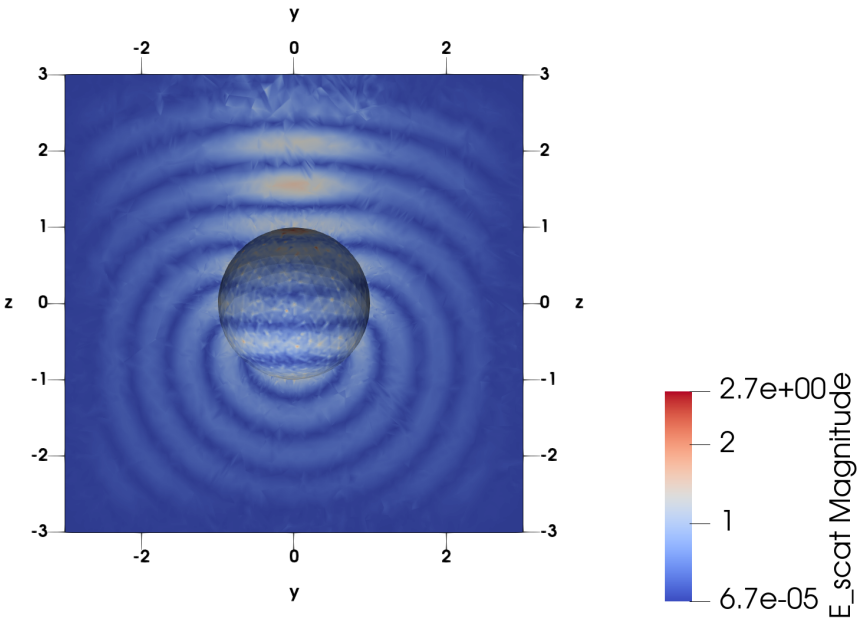


Figure 6: Scattered-field visualization from the MN5 EPW4/P4 run. The outgoing pattern around the sphere is the field used by the near-to-far RCS extraction after the time-windowed Fourier transform.

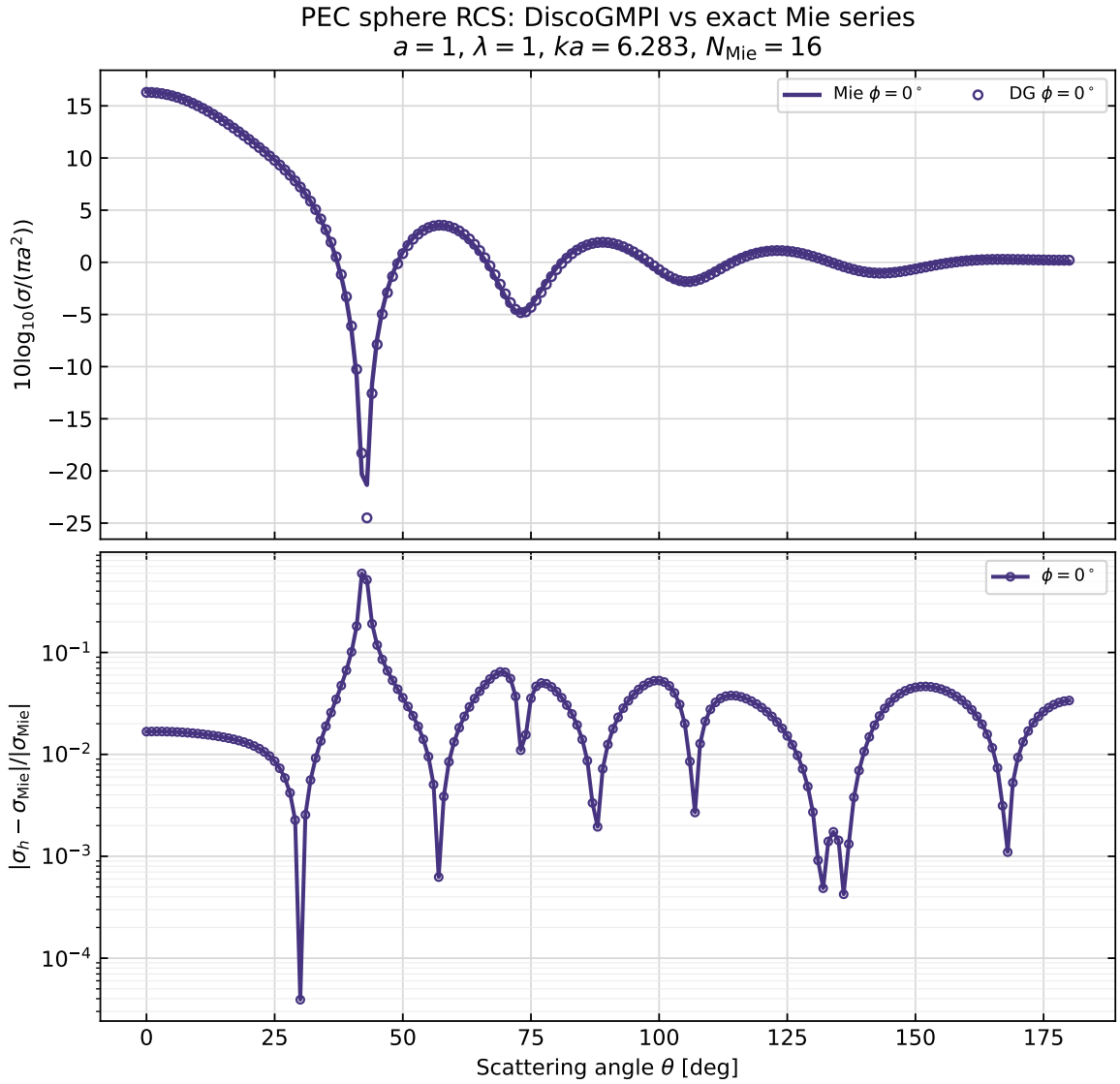


Figure 7: MN5 EPW4/P4 Cartesian RCS comparison. The upper panel compares the DG near-to-far result with the exact Mie curve for $10\log_{10}(\sigma/(\pi a^2))$. The lower panel reports the relative error; the visible peaks correspond mainly to angular neighborhoods of Mie nulls.

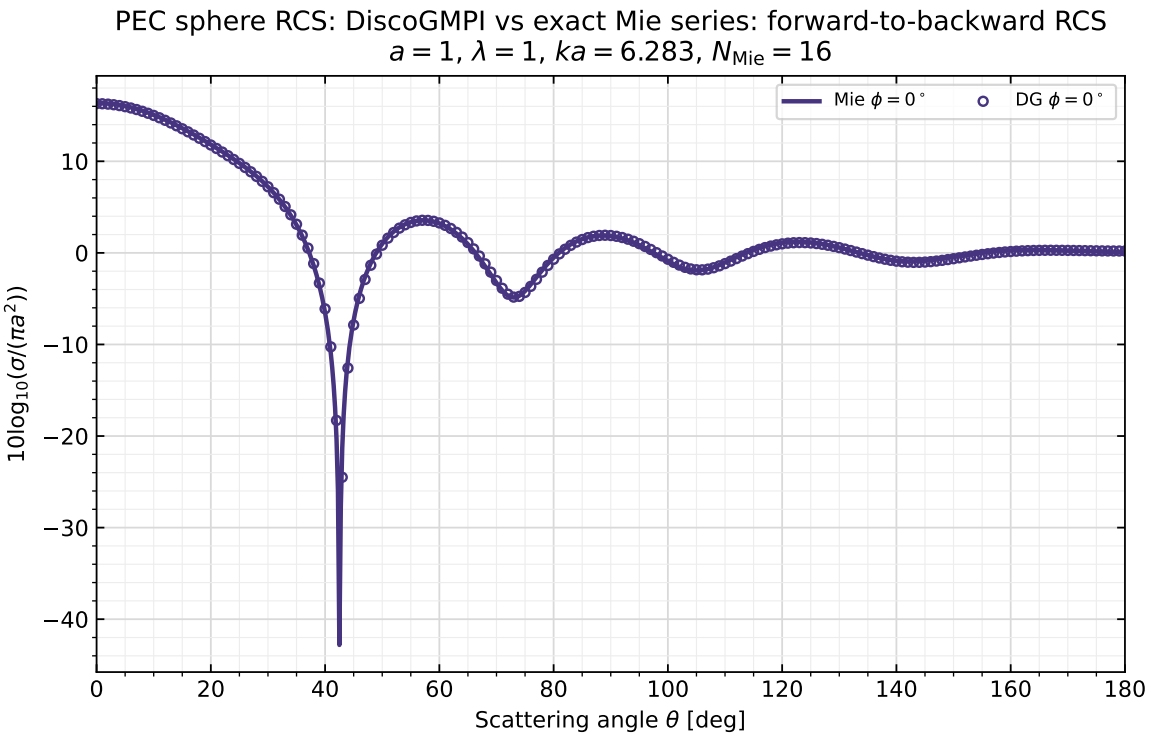


Figure 8: Forward-to-backward RCS cut for the MN5 EPW4/P4 run. Here $\theta = 0°$ is the forward $+z$ direction and $\theta = 180°$ is the backscatter direction.

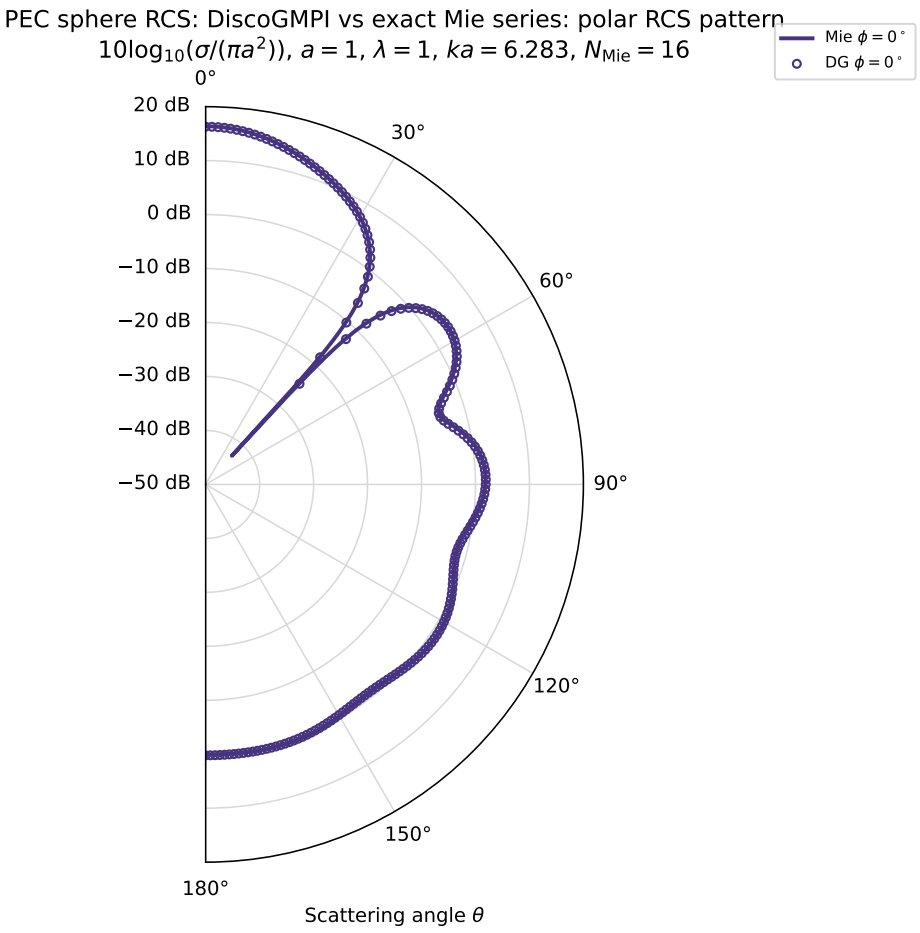


Figure 9: Polar representation of the MN5 EPW4/P4 RCS pattern. The radial coordinate is the normalized dB RCS, $10\log_{10}(\sigma/(\pi a^2))$.

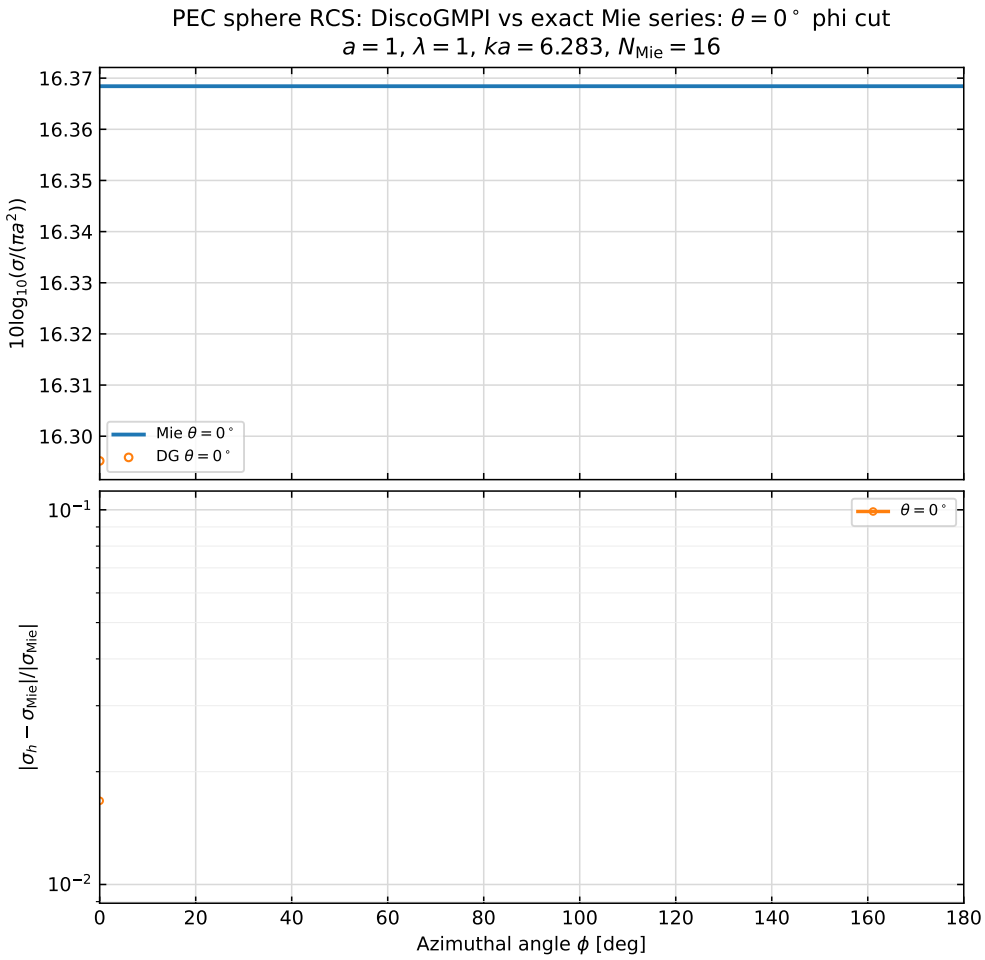


Figure 10: $\theta = 0°$ azimuthal diagnostic produced by the MN5 EPW4/P4 postprocessor. This run requested only $\phi = 0°$, so the DG data contain one azimuth sample. The figure is therefore a bookkeeping and orientation check, not a full ϕ-sweep validation.

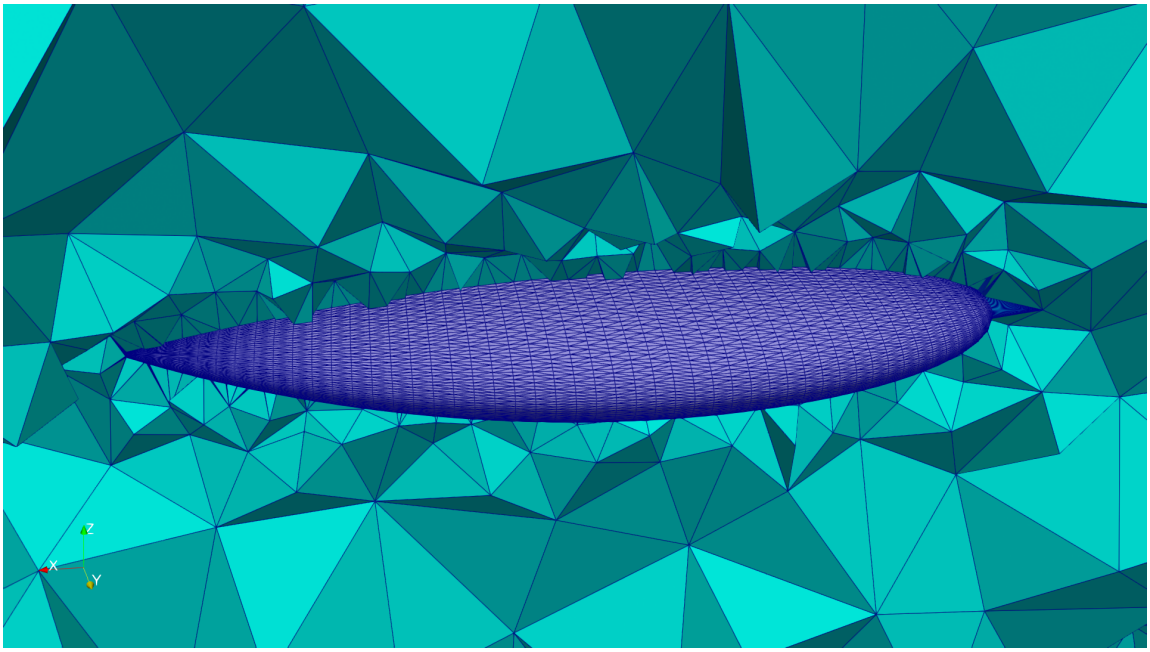


Figure 11: NASA PEC almond volume-mesh view from the Austin benchmark mesh folder. The almond is embedded as an inner PEC boundary in a tetrahedral volume mesh. This view is useful for inspecting near-target refinement, surrounding air volume, and the transition to coarser elements away from the scatterer.

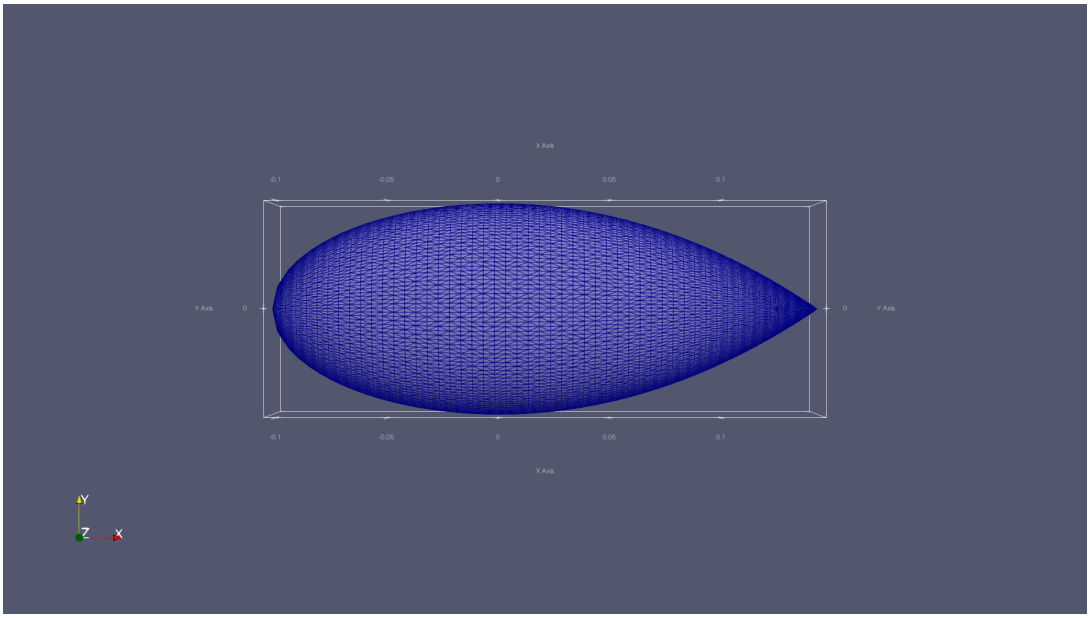


Figure 12: xy-plane PNG view of the Austin PEC Almond benchmark mesh. This planform view shows the rounded nose and sharp downstream tip, which are the main geometric features that drive local refinement and timestep cost.

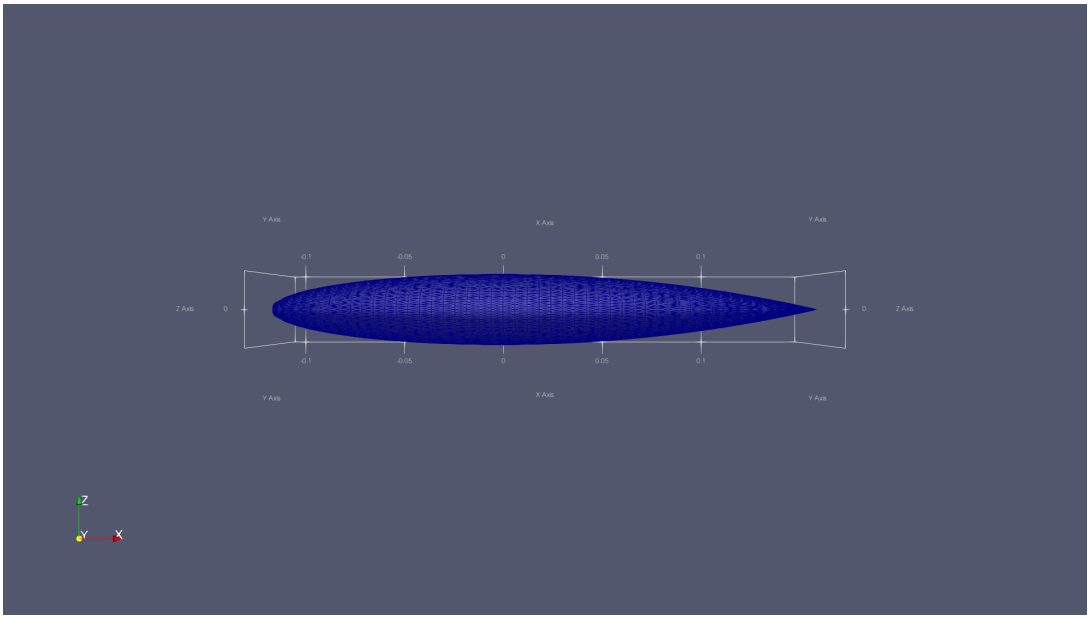


Figure 13: xz-plane PNG view of the Austin PEC Almond benchmark mesh. This longitudinal view shows the slender body profile and vertical thickness that must be resolved by the near-surface tetrahedra.

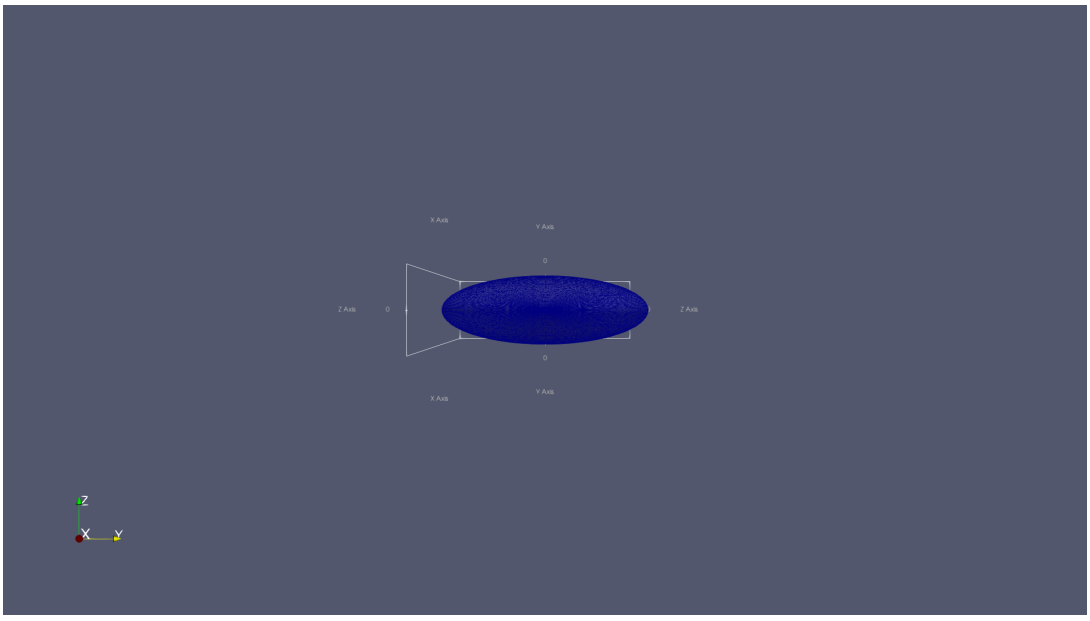


Figure 14: yz-plane PNG view of the Austin PEC Almond benchmark mesh. This transverse section is useful for checking the thin almond thickness and for identifying cross-sectional mesh-quality issues before a long RCS run.